

TCP/IP Model – Detailed Notes

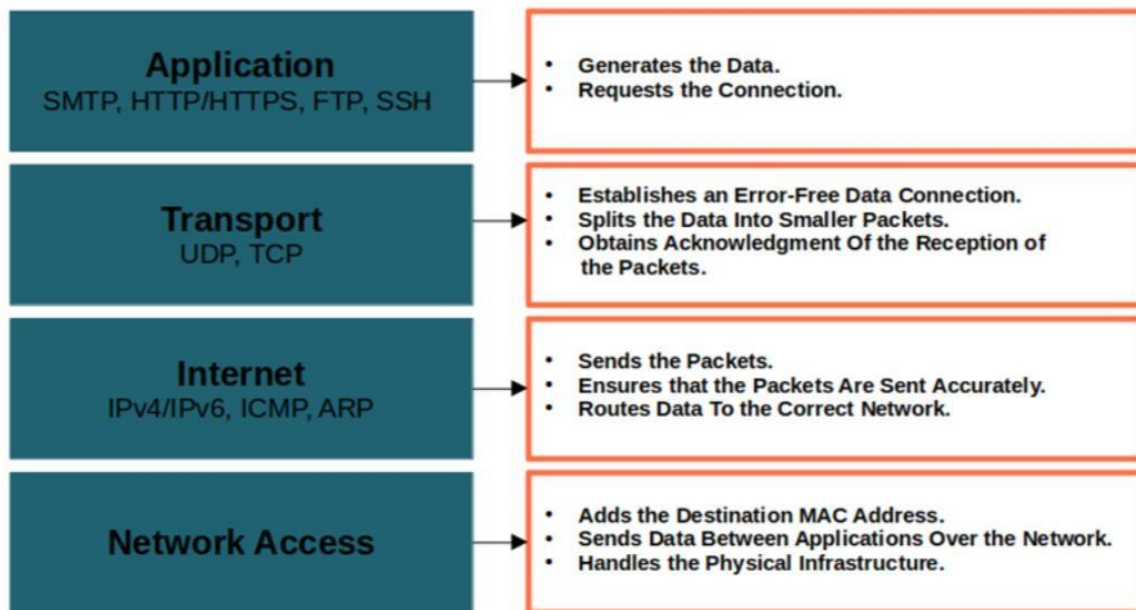
1. Introduction

The **TCP/IP model** stands for **Transmission Control Protocol / Internet Protocol**. It is the **core communication model of the Internet** and defines how data is transmitted between computers.

The model provides a **set of rules (protocols)** that allow devices with different hardware and software to communicate seamlessly over networks.

Developed by the **U.S. Department of Defense (DoD)** during the **1970s** as part of the **ARPANET** project, the TCP/IP model became the **foundation of modern networking and the Internet**.

The Four Layers Of the TCP/IP Model and Their Functions



2. Purpose of the TCP/IP Model

The main purposes of the TCP/IP model are:

- To provide a **standardized communication framework** for devices across the world.

- To define **how data is addressed, transmitted, routed, and received**.
- To enable **interoperability** between different computer systems and networks.
- To ensure **reliable data transfer** even when networks are complex or unreliable.

3. Layers of the TCP/IP Model

The TCP/IP model consists of **four layers**, each performing specific network functions. Every layer depends on the layer below it and supports the layer above it.

1. Network Interface Layer (Link Layer)

Functions:

- Responsible for the **physical transmission of data** across the network.
- Manages **hardware addressing (MAC addresses)** and **error detection**.
- Deals with how bits are physically placed on the transmission medium (cables, radio waves, etc.).
- Works closely with **network hardware devices** such as switches, routers, and NICs.

Key Concepts:

- Converts packets from the Internet Layer into **frames** for physical transmission.
- Ensures data is transmitted without hardware errors.

Common Protocols:

- **Ethernet** – used for LAN connections.
- **Wi-Fi (IEEE 802.11)** – used for wireless communication.
- **ARP (Address Resolution Protocol)** – maps IP addresses to MAC addresses.
- **PPP (Point-to-Point Protocol)** – used in direct connections between two nodes.

2. Internet Layer

Functions: • Handles **logical addressing and routing** of

data.

- Defines how data packets travel from the source network to the destination network.
- Determines the **best path** for data through interconnected networks.
- Responsible for **packet fragmentation** and **reassembly** if needed.

Key Concepts:

- Each device connected to a network has a unique **IP address**.
- Data is transmitted in small units called **packets** or **datagrams**.
- Routers operate at this layer to direct packets to their destinations.

Common Protocols:

- **IP (Internet Protocol)** – Core protocol for addressing and routing.
 - **IPv4:** Uses 32-bit addressing.
 - **IPv6:** Uses 128-bit addressing for more devices.
- **ICMP (Internet Control Message Protocol):** Used for error messages and diagnostics (e.g., ping).
- **IGMP (Internet Group Management Protocol):** Used for multicast communication.

3. Transport Layer

Functions:

- Provides **end-to-end communication** between applications on different devices.
- Ensures **data reliability, sequencing, and flow control**.
- Breaks large messages into smaller segments for easier transmission.
- Manages error detection and retransmission of lost data.

Key Concepts:

- Responsible for the **logical connection** between sender and receiver.
- Provides **data integrity** and **flow management**.
- Can operate in **connection-oriented** or **connectionless** modes.

Common Protocols:

- **TCP (Transmission Control Protocol):**
 - Connection-oriented. ○ Ensures reliable delivery using acknowledgments and retransmissions.
 - Used in applications like HTTP, FTP, SMTP, and Telnet.
- **UDP (User Datagram Protocol):** ○ Connectionless. ○ Faster but less reliable. ○ Used in real-time services like DNS, video streaming, and VoIP.

4. Application Layer

Functions:

- Provides **services directly to end users** and applications.
- Defines **how network applications** (like browsers, email clients, or file transfer tools) communicate.
- Responsible for **data formatting, encryption, and session management**.

Key Concepts: • The **interface between user applications and the network**.

- Each protocol in this layer serves a specific purpose such as web browsing, email, or file sharing.

Common Protocols:

- **HTTP / HTTPS:** Web browsing (transfers web pages).
- **FTP / SFTP:** File transfer between computers.
- **SMTP / POP3 / IMAP:** Sending and receiving emails.
- **DNS:** Converts domain names to IP addresses.
- **SSH / Telnet:** Remote access to computers.
- **SNMP:** Network management and monitoring.

Why TCP/IP is used over the OSI model -

